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RESEARCH ARTICLE

A Multitask Learning-Based Vision Transformer for Plant Disease Localization and Classification

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Abstract
Plant disease detection is a critical task in agriculture, essential for ensuring crop health and productivity. Traditional methods in this context are often labor-intensive and prone to errors, highlighting the need for automated solutions. While computer vision-based solutions have been successfully deployed in recent years for plant disease identification and localization tasks, these often operate independently, leading to suboptimal performance. It is essential to develop an integrated solution combining these two tasks for improved efficiency and accuracy. This research proposes the innovative Plant Disease Localization and Classification model based on Vision Transformer (PDLC-ViT), which integrates co-scale, co-attention, and cross-attention mechanisms and a ViT, within a Multi-Task Learning (MTL) framework. The model was trained and evaluated on the Plant Village dataset. Key hyperparameters, including learning rate, batch size, dropout ratio, and regularization factor, were optimized through a thorough grid search. Early stopping based on validation loss was employed to prevent overfitting. The PDLC-ViT model demonstrated significant improvements in plant disease localization and classification tasks. The integration of co-scale, co-attention, and cross-attention mechanisms allowed the model to capture multi-scale dependencies and enhance feature learning, leading to superior performance compared to existing models. The PDLC-ViT model evaluated on two public datasets achieved an accuracy of 99.97%, a Mean Average Precision (MAP) of 99.18%, and a Mean Average Recall (MAR) of 99.11%. These results underscore the model's exceptional precision and recall, highlighting its robustness and reliability in detecting and classifying plant diseases. The PDLC-ViT model sets a new benchmark in plant disease detection, offering a reliable and advanced tool for agricultural applications. Its ability to integrate localization and classification tasks within an MTL framework promotes timely and accurate disease management, contributing to sustainable agriculture and food security.

Keywords Plant disease · Classification · Localization · Vision transformer · Multi-task learning

Abbreviations

PLDC-ViT	Plant Disease Localization and Classification Vision Transformer
MAP	Mean Average Precision
MAR	Mean Average Recall
AUC	Area Under the Curve
CNN	Convolutional Neural Network
ViT	Vision Transformer
MTL	Multi-Task Learning
XAI	Explainable Artificial Intelligence
Grad-CAM	Gradient-weighted Class Activation Mapping
GAM	Gradient Activation Map
IoU	Intersection over Union
ACC	Accuracy
Sn	Sensitivity (Recall)
Sp	Specificity
Pre	Precision
F1	F1 Score
MSCVT	Multiscale Convolution and Vision Transformer
CMTL	Concurrent Multi-Task Learning

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1 Introduction

Plant diseases can have a severe impact on crop health, leading to substantial damage, reduced yields, and financial losses for farmers. Addressing these diseases is of utmost importance to ensure food security and sustainable agriculture. These diseases encompass various infections, pathogens, and abnormalities that adversely affect the health and growth of plants. They can be caused by bacteria, fungi, viruses, nematodes, or environmental factors such as nutrient deficiencies or adverse weather conditions [1]. The timely and precise detection of plant diseases plays a vital role in implementing effective disease management and prevention strategies.

Conventional plant disease detection methods heavily depend on manual visual inspection conducted by experts. However, these methods have several drawbacks, including being time-consuming, subjective, and susceptible to human errors. In recent years, automated plant disease detection systems have shown great promise in overcoming the limitations of traditional methods [2]. In recent times, there have been significant breakthroughs in deep learning and computer vision, presenting promising opportunities for automating the detection and classification of plant diseases [3]. These cutting-edge approaches have transformed the field of plant disease detection, harnessing the capabilities of artificial neural networks to extract meaningful features from plant images automatically. This advanced technology enables accurate and efficient disease classification and detection, revolutionizing the way plant diseases are diagnosed using images.

Convolutional Neural Networks (CNNs) [4] have been widely employed for plant disease detection due to their ability to capture spatial dependencies in images. These networks are built with multiple layers of convolutional and pooling operations, allowing them to learn hierarchical representations of input images. CNNs have shown remarkable performance in various plant disease detection tasks, achieving high accuracy and robustness.

Transfer learning is a commonly used technique in deep learning-based plant disease detection [5]. Pretrained CNN models, such as VGG [6], ResNet [7], or Inception [8], trained on large-scale image datasets are utilized as feature extractors. Subsequently, the pretrained models are fine-tuned on plant disease datasets, allowing the learned features to be tailored and optimized for disease detection tasks. Transfer learning accelerates the training process, especially when the available plant disease dataset is limited.

CNNs, although widely used in computer vision, have limitations in capturing long-range dependencies and rely heavily on large labeled datasets, making it challenging to capture global context information needed for specific plant disease detection scenarios. Limited or imbalanced datasets can result in overfitting or biased models, impacting real-world performance. Additionally, CNNs can be computationally expensive, especially with deeper architectures, necessitating significant computational resources and training time. Similarly, ResNet architectures encounter limitations such as the vanishing gradient problem with increasing layers, increased memory requirements for deeper models, and the potential for overfitting due to added parameters from skip connections. Inception models, known for their complexity, can suffer from information loss through spatial downsampling, affecting the retention of fine-grained details crucial for accurate disease detection. They also demand more computational resources and longer training time while limiting interpretability and transparency. The limitations of conventional deep learning models used for plant disease detection are extensively discussed by Arsenovic et al. [9].

While deep learning models have achieved remarkable results, they often focus on either disease localization or disease classification as separate tasks. Integrating both localization and classification tasks into a multitasking framework can provide a more comprehensive and practical solution for plant disease detection. Empirical studies have demonstrated that MultiTask Learning (MTL), which involves the simultaneous learning of features from multiple related tasks, offers significant advantages in prediction performance compared to discrete tasks [10]. MTL mitigates the risk of overfitting in individual tasks by training them in parallel, enabling the model to utilize a broader range of features from different tasks. This results in improved generalization capabilities, enhancing the overall performance of the model.

ViTs are a recent breakthrough in computer vision, adapting the Transformer architecture, originally designed for natural language processing, to handle image-based tasks [11]. Unlike CNNs, which rely on convolutional operations for feature extraction, ViTs adopt a self-attention mechanism to capture global dependencies within an image. This allows ViTs to model relationships between all image patches, enabling them to learn meaningful representations and achieve state-of-the-art performance in various computer vision tasks. Significant works on MTL models and ViTs for plant disease detection have been proposed in [12] and [13], respectively.

This investigation stems from the need for innovative technology driven solutions for sustainable agriculture, and the limitations of existing models. These models often fail to effectively handle multiple tasks simultaneously, leading to suboptimal performance in both disease localization and classification. The integration of these two tasks is crucial for sustainable agriculture as it enables precise, targeted interventions, reducing the need for broad-spectrum pesticides and minimizing environmental impact. This approach can facilitate timely management of plant diseases, thereby enhancing crop health and productivity. Inspired by the demand for such integral models, and the demonstrated abilities of MTL frameworks and ViTs, this research aims to build the PLDC-ViT for plant disease localization and classification. The contributions of this research are as below:

1. ViT-based MTL framework leveraging the power of ViTs to capture global dependencies and encode local and global information from plant images.
2. Joint localization and classification by jointly considering disease localization and classification as interconnected tasks for a comprehensive and accurate analysis of plant diseases, leading to improved performance and more effective disease management.

The rest of this paper is structured as follows: Sect. 2 provides a review of related works in this research domain. Section 3 describes the datasets employed and methods used in this study. In Sect. 4, the proposed problem, the PDLC-ViT framework, and the training procedure are discussed. Section 5 presents the empirical results, comparative analyses, and the key findings of the study. Finally, Sect. 6 concludes the paper by summarizing the contributions, highlighting the research's significance, potential applications, and opportunities for future extensions.

2 Related Works

This section offers an extensive overview of prior works in the context of this research, encompassing deep learning models, MTL-based works, and ViTs, with the goal of identifying their strengths, limitations, and research gaps, thus forming the basis for the proposed method. An extensive review by Li et al. [14] presents the recent research progress in using deep learning technology for crop leaf disease identification. The authors discuss the current trends, challenges, and unresolved issues pertaining to plant leaf disease detection using deep learning and advanced imaging techniques are discussed, with the aim of providing a valuable resource for researchers in the field.

Deep learning techniques were initially applied to plant image recognition with a focus on analyzing leaf vein patterns. A notable study employed a CNN architecture with 3–6 layers to successfully classify three leguminous plant species: white bean, red bean, and soybean [15]. The CNN model extracted and analyzed intricate features of leaf vein structures unique to each species, capturing low-level features such as edges and textures and progressing to more complex patterns. This hierarchical feature extraction enabled effective species differentiation based on leaf vein characteristics. The study demonstrated the potential of CNNs in plant species classification, highlighting the effectiveness of deep learning in capturing subtle biological variations for accurate identification, and laying the groundwork for further applications in plant image recognition.

Subsequently, Mohanty et al. [16] employed a classifier model to accurately identify 14 different crop species and 26 crop diseases, achieving an impressive accuracy of 99.35%. Building upon this work, Kawasaki et al. [17] proposed a CNN-based system specifically designed for the precise recognition of diseases affecting cucumber leaves, achieving a commendable accuracy rate of 94.9%. In an investigation by Ma et al. [18], a deep CNN was employed for the identification and recognition of four diseases in cucumber plants, including downy mildew, anthracnose, powdery mildew, and target leaf spots, based on their distinct symptoms. This study reported a recognition accuracy of 93.4%.

Impressive strides have been made in plant leaf disease recognition; however, the availability of diverse datasets remains limited. CNN training necessitates extensive and diverse datasets, which are currently scarce in the field of plant leaf disease recognition. Consequently, transfer learning has emerged as a viable solution to enhance the robustness of CNN classifiers. By adapting pre-trained CNNs and retraining them with smaller datasets showcasing different distributions, transfer learning has proven to be an effective approach in implementing deep learning models [19]. Promising results have been demonstrated by leveraging pre-trained CNN models from the ImageNet dataset and retraining them for leaf disease recognition [20].

Recently, MTL models have been explored in the field of plant disease detection, enabling the simultaneous execution of multiple tasks. These models can effectively handle tasks, such as severity estimation of leaf disease, identification of plant species, and classification of plant disease concurrently [21]. To address the limitations in information utilization and scalability of existing models, Lee et al. [22] introduced a novel Conditional Multi-Task Learning (CMTL) model in this context. This method employs an MTL mechanism with a conditional scheme that links host species and disease representations, mimicking how plant pathologists infer diseases from species information. This improves plant disease identification performance compared to traditional joint species–disease pair modeling. It efficiently utilizes available information and is scalable, not requiring distinctions between similar species or diseases, thus enhancing accuracy and suitability for real-world applications.

However, ViTs offer advantages over MTL frameworks in plant disease detection. They capture both local and global context information, enabling accurate analysis of plant images. ViTs also generalize well with limited labeled data through pre-training on large-scale datasets like ImageNet. Additionally, they provide flexibility and scalability by adapting to different disease detection tasks with a single model, reducing complexity and computational requirements.

In their work, Thakur et al. [13] introduced a hybrid model, named PlantViT, for automating plant disease detection, which integrates the strengths of a CNN and a ViT, incorporating a multi-head attention module. The model was evaluated on two extensive plant disease detection datasets, namely, PlantVillage and Embrapa. Experimental results showed that the model achieves detection accuracies of 98.61% and 87.87% on the PlantVillage and the Embrapa datasets, respectively. These results outperform the current state-of-the-art methods, highlighting the effectiveness of PlantViT in plant disease detection.

In [23], a lightweight deep learning approach leveraging the ViT architecture is proposed for real-time automated plant disease classification. The study rigorously compares various models, including ViT, traditional CNNs, and a hybrid of CNN and ViT. Extensive training and evaluation across multiple datasets reveal that while attention blocks significantly enhance accuracy, they also introduce latency in predictions. However, integrating attention blocks with CNN blocks strikes an optimal balance between accuracy and speed, offering a robust solution for real-time plant disease classification.

In recent studies, deep learning models have been employed to identify cassava leaf diseases, replacing the traditional and unreliable intuitive diagnosis conducted by farmers. To improve the disease detection accuracy in these species, Thai et al. [24] explored the use of the ViT model instead of a CNN. Experimental results demonstrate that the ViT model achieves competitive accuracy, surpassing popular CNN models such as EfficientNet and Resnet50 on the Cassava Leaf Disease Dataset. These findings highlight the potential superiority of the ViT model in analyzing leaf diseases.

An extended study by Thakur et al. [25] addressed the underexplored application of ViTs in plant pathology by introducing PlantXViT, a light-weight model based on CNNs with ViTs for efficient identification of various plant diseases across different crops, well suited for IoT-based smart agriculture. Evaluation on five datasets shows that PlantXViT outperforms state-of-the-art models, achieving average accuracies of over 93.55%, 92.59%, and 98.33% on Apple, Maize, and Rice datasets, respectively, even in challenging backgrounds. The model's explainability is also assessed using gradient-weighted class activation maps, demonstrating its ability to provide insights into the decision-making process.

Zhu et al. [26] introduced the Multiscale Convolution and Vision Transformer (MSCVT), a hybrid model that integrates multiscale convolutions with ViT techniques for precise crop disease identification. This approach combines the strengths of multiscale convolutions, which capture fine-grained local features, with ViT's ability to model long-range dependencies and global context. The study demonstrates that MSCVT outperforms single-scale CNNs and standalone ViT models by effectively capturing local and global details while optimizing computational efficiency. This balance ensures high accuracy in disease identification and practical deployment in resource-constrained environments, making MSCVT a valuable tool for early detection and management of crop diseases.

The ConvViT [27] is a lightweight ViT-based approach developed for apple leaf disease identification. It integrates convolutional and Transformer structures to effectively capture both global and local features of crop disease spots, enhancing overall robustness and accuracy. The patch embedding method is improved to preserve edge information and facilitate information exchange within the Transformer. By leveraging depthwise separable convolution and linear-complexity multi-head attention operations, ConvViT achieves substantial reduction in resource requirements. This model achieves comparable identification performance (96.85%) on a complex background apple leaf disease dataset, with significantly reduced parameters (32.7%) and FLOPs (21.7%). This highlights the effectiveness and practicality of ConvViT as a disease identification model.

The Shuffle Convolution-based Lightweight Vision Transformer (SLViT) [28] is a hybrid network designed for the accurate diagnosis of sugarcane leaf diseases. The study demonstrates that SLViT, which combines a lightweight CNN architecture with a flexible plug-in Transformer encoder, surpasses the state-of-the-art models on the publicly available Plant Village dataset with respect to the metrics speed, weight, memory usage, and precision.

The effectiveness of MTL in enhancing correlated tasks has been well established. Existing approaches typically employ a backbone for initial feature extraction and utilize exclusive branches for each task. However, the sharing of information between branches is often accomplished through basic concatenation or summation of feature maps, which results in overlooking the local image features and neglecting the significance or correlation between tasks.

MTL-based ViTs offer a significant advancement in addressing the limitations of existing approaches. By integrating ViTs into the MTL framework, these models can capture both the global and local characteristics of images, enabling more comprehensive information exchange between tasks. This integration allows for a better understanding of the relationships and dependencies among tasks, leading to improved performance across multiple related tasks. MTL-based ViTs provide a more holistic approach to leveraging shared knowledge and extracting meaningful features, resulting in enhanced accuracy and robustness in correlated task scenarios.

In the recent years, ViTs have been used in performing multiple tasks simultaneously sharing the data across different tasks. The Unified Transformer (UniT) [29] model is proposed for the simultaneous learning of multiple tasks across diverse domains, encompassing object detection, natural language understanding, and multimodal reasoning. The UniT encodes input modalities using an encoder and generates task predictions using a shared decoder over the encoded input representations. Task-specific output heads are employed, and the entire model is trained jointly end-to-end with task losses. UniT distinguishes itself by employing shared model parameters across all tasks, accommodating a wide range of tasks across domains, and delivering robust performance across 7 tasks on 8 datasets with reduced parameter count.

Similarly, the MulT [30] framework proposes an end-to-end MTL-based Transformer for simultaneous learning of multiple high-level vision tasks. It outperforms state-of-the-art multitask CNNs and single-task Transformers, demonstrating the significance of shared attention across tasks. The MulT model shows robustness and generalization to new domains in multitask benchmarks. The study highlights the advantages of Transformer-based architectures for MTL, such as better scalability, improved feature representation, and the ability to capture complex dependencies between tasks, thereby providing a substantial contribution to the field and paving the way for future research and applications in domains requiring robust multitask learning solutions.

To address the challenges of constructing a comprehensive dataset for diagnosing and quantifying the COVID-19 severity using Chest X-ray (CXR) data, a novel Multi-task ViT architecture is proposed by Park et al. [31]. This approach leverages abundant unlabeled CXR data through self-attention modeling, overcoming the limitations of existing ViT models that are not optimized for CXR. By utilizing a backbone network trained on public datasets, the multi-task ViT extracts informative low-level features from the images, which serve as valuable resources for a robust ViT. Best performances are exhibited by the model in both diagnosis and severity quantification tasks, and remarkable generalization capability is demonstrated on external test datasets from diverse institutions. These promising results underscore the potential for widespread deployment of the model in the medical field.

Tian et al. [32] introduced the MultiTask ViT (MTViT), a representation learning method that leverages ViTs. MTViT introduces a multiple branch transformer that sequentially processes image patches associated with different tasks, allowing for information exchange through the Cross-Task attention (CA) module. Unlike previous models, MTViT utilizes ViT's self-attention mechanism for feature extraction, resulting in improved memory and computation efficiency. Experimental results on benchmark datasets demonstrate that MTViT outperforms or achieves comparable performance to existing CNN-based MTL methods.

Despite the widespread adoption of MTL-based ViTs in various computer vision applications, there remains a noticeable research gap in their application to plant disease detection. The work proposed by Goncalves et al. [33], known as MTLSegFormer, stands out as the only existing effort in exploring the potential of MTL-based ViTs to address plant disease detection challenges. MTLSegFormer is a semantic segmentation model designed for precision agriculture that leverages both MTL and attention mechanisms. It combines MTL and attention mechanisms, learning two feature maps for each task after extracting backbone features. The model assigns weights to important local regions for specific tasks, resulting in improved accuracy for correlated tasks in precision agriculture.

This review highlights the significance of ViT-based models in plant disease detection, demonstrating their effectiveness in accurately identifying and localizing plant diseases. However, there is a notable lack of MTL-based ViT models in this domain. Integrating MTL with ViT could enhance performance by simultaneously handling related tasks, such as disease classification and localization, within a unified framework. This research addresses this gap by developing an MTL-based ViT model, aiming to improve accuracy, efficiency, and scalability in plant disease detection. By leveraging the strengths of MTL and ViT architectures, this study provides a robust solution that significantly contributes to sustainable agricultural practices and food security.

3 Materials and Methods

This section provides a comprehensive overview of the dataset used, including its source, composition, and preprocessing steps. It also outlines the implementation details, covering aspects such as the programming language, libraries or frameworks employed, and hardware specifications. These details establish the foundation for understanding the methodology and experimental setup of the PDLC-ViT model, ensuring transparency and reproducibility.

3.1 Dataset

The PDLC-ViT is trained and tested with the PlantVillage [34] and the PlantDoc [35] datasets. The PlantVillage dataset contains 54,306 healthy and unhealthy leaf images categorized into 38 different classes based on species and disease. It was curated by researchers at Penn State University to support the advancement of mobile disease diagnostics through machine vision approaches. The public dataset provides researchers and developers with a valuable resource for exploring and evaluating machine learning techniques in automated plant disease detection and classification in agriculture. The distribution of the plant classes for training and testing is given in Table 1. The original dataset is augmented with images by affine transformation from the original images. The Plantdoc dataset consists of an augmented dataset of 2940 images collected from 13 species with corresponding segmentation masks. In this research, this dataset is used to evaluate the disease localization ability of the model.

Table 1 Distribution of classes in plant village dataset

Leaf	Infection	Training	Testing
Apple	Scab	630	630
	Black_rot	621	621
	Rust	275	275
	Healthy	1645	1645
Blackberry	Healthy	1502	1502
Cherry	Healthy	854	854
	Powdery_mildew	1052	1052
Corn_(maize)	Gray_leaf_spot	513	513
	Common_rust	1192	1192
	Healthy	1162	1162
	Leaf Blight	985	985
Grape	Black_rot	1180	1180
	Black_Measles	1384	1384
	Healthy	423	423
	Leaf Blight	1076	1076
Citrus	Greening	5507	5507
Peach	Bacterial_Spot	2297	2297
	Healthy	360	360
Pepper_Bell	Bacterial_Spot	997	997
	Healthy	1478	1478
Potato	Early Blight	1000	1000
	Healthy	152	152
	Late Blight	1000	1000
Raspberry	Healthy	371	371
Soybean	Healthy	5090	5090
Squash	Powdery_mildew	1835	1835
Strawberry	Healthy	456	456
	Leaf_scorch	1109	1109
Tomato	Bacterial_Spot	2127	2127
	Early_Blight	1000	1000
	Healthy	1591	1591
	Late Blight	1909	1909
	Leaf Mold	952	952
	Leaf Spot	1771	1771
	Spider Mite	1676	1676
	Target Spot	1404	1404
	Mosaic Virus	373	373
	Leaf Curl Virus	5357	5357
Total		54,306	54,306

3.2 Details of Implementation

The PDLC-ViT is implemented using Matlab 2023a with deep learning and image processing toolboxes. The training and testing occur on a high-performance system with an i7 processor powered by 16 GB DDR4 RAM, and NVIDIA GeForce RTX 3070 8 GB Graphics card for efficient computation. This setup ensures accurate plant disease detection and classification performance assessment.

4 Proposed PDLC-ViT Model

This section gives a mathematical description of the joint segmentation and classification problem addressed in this research, detailed architecture of the proposed model and its components and the training procedure.

4.1 Problem Formulation

Given a dataset $D = (x_i, y_i)_{i=1}^{N}$, where x_i represents the ith input image and y_i represents the corresponding ground truth label, the problem of disease localization and classification in plant leaf images can be formulated as follows. Let $X \in \mathbb{R}^{H\times W\times C}$ represent the input image tensor, where H, W, and C denote the height, width, and number of channels, respectively. The disease localization task aims to predict a binary mask $M \in \{0,1\}^{H\times W}$ that indicates the presence or absence of the disease in each pixel of the image. The disease classification task involves predicting the class label $L \in 1,2,\ldots,K$, where K is the total number of disease classes.

To jointly address the disease localization and classification tasks, an MTL model is proposed. This model consists of a shared feature extractor f_{shared} and task-specific modules, f_{loc} for localization and f_{cls} for classification. The shared feature extractor maps the input image X to a shared feature representation $Z \in \mathbb{R}^{H'\times W'\times d}$, where H', W', and d represent the dimensions of the feature map. The localization module takes the shared feature representation Z and produces the disease mask M using a mapping function $f_{\text{loc}} : \mathbb{R}^{H'\times W'\times d} \to \{0,1\}^{H\times W}$. Similarly, the classification module utilizes the shared features Z and predicts the disease class label L using a mapping function $f_{\text{cls}} : \mathbb{R}^{H'\times W'\times d} \to \{1,2,\ldots,K\}$.

The MTL model is trained by minimizing a combination of loss functions for both tasks. The disease localization loss L_{loc} measures the pixel-wise discrepancy between the predicted mask M and the ground truth mask. This can be expressed as in (1), where $\mathcal{L}_{\text{loc}}$ is a loss function and M_{gt} denotes the ground truth mask

$$L_{\mathrm{loc}} = \frac{1}{HW}\sum_{i=1}^{H}\sum_{j=1}^{W}\mathcal{L}_{\mathrm{loc}}(M_{ij}, M_{\mathrm{gt}_{ij}}). \tag{1}$$

The disease classification loss L_{cls} measures the discrepancy between the predicted class label L and the ground truth label y. This can be expressed as in (2) where $\mathcal{L}_{\mathrm{cls}}$ is the categorical cross-entropy loss

$$L_{\mathrm{cls}} = \mathcal{L}_{\mathrm{cls}}(L, y). \tag{2}$$

The overall loss function for the MTL model is defined as in (3), where λ_{loc} and λ_{cls} are hyperparameters that control the importance of each task during training. The MTL model is trained by minimizing the loss using the Adam optimization algorithm. By jointly optimizing the disease localization and classification tasks, the proposed MTL model aims to improve the accuracy and efficiency of disease detection in plant leaf images, leveraging the shared information between the tasks for enhanced performance

$$L_{\mathrm{MTL}} = \lambda_{\mathrm{loc}}L_{\mathrm{loc}} + \lambda_{\mathrm{cls}}L_{\mathrm{cls}}. \tag{3}$$

4.2 PDLC-ViT High-Level Architecture

The PDLC-ViT is implemented with two parallel branches, one each for segmentation and classification, each comprising the co-scale and co-attention layers, respectively. The co-scale layer is strategically selected to capture multi-scale dependencies, which are pivotal for accurately detecting disease patterns that can vary significantly in size. This layer ensures robust feature extraction, enhancing the model's ability to generalize across different plant diseases. The co-attention layer further refines the representation of image patches by emphasizing spatial relationships, effectively integrating both local details and global context. This dual focus is crucial for precise disease localization and classification. Further, the cross-attention mechanism serves as a sophisticated integrator, facilitating seamless information exchange between the segmentation and classification tasks. By combining these enriched representations, the model achieves a comprehensive understanding of the input image, significantly improving overall performance. This integrated, multi-faceted approach ensures a holistic analysis, leading to more accurate, efficient, and reliable plant disease detection.

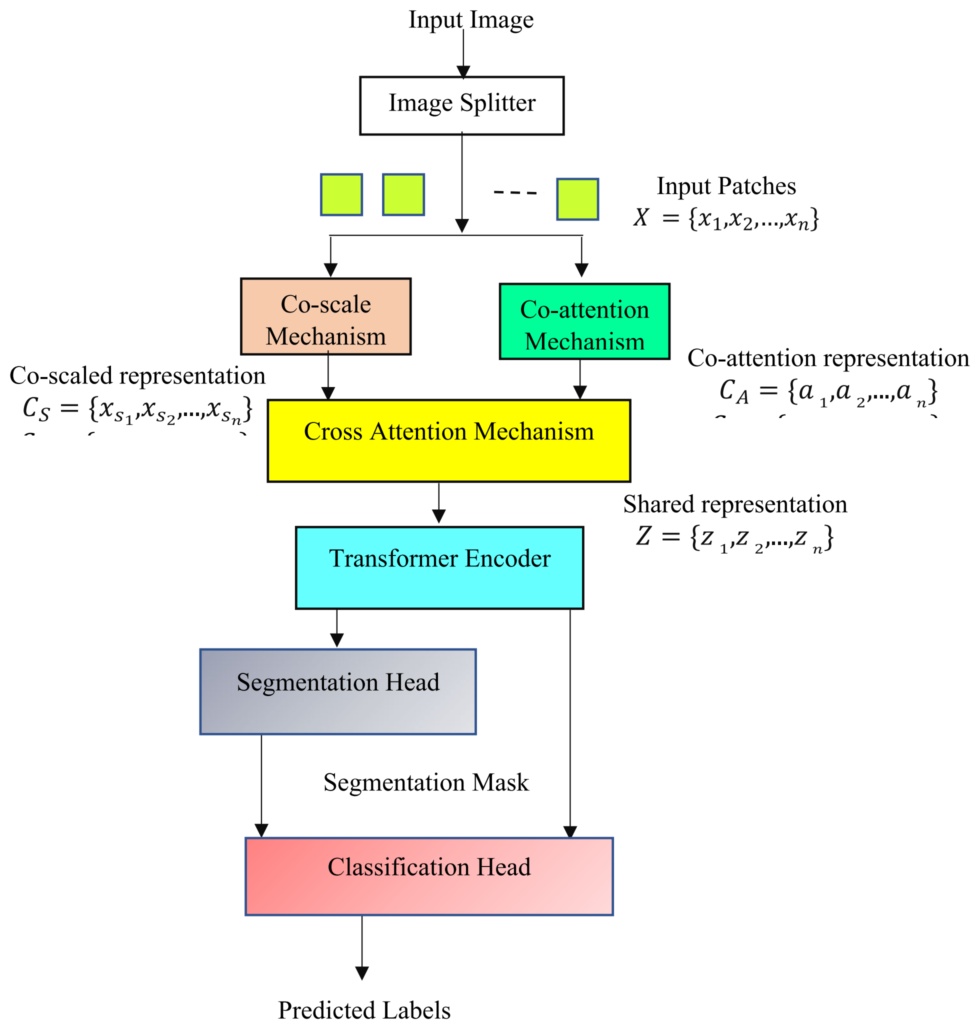
Given an input image I to a patch splitter, it is divided into a set of non-overlapping patches $X = \{x_1, x_2, \ldots, x_n\}$. The co-scale layer captures dependencies and interactions at different spatial scales between the patches and produces the output $C_S = \{x_{s_1}, x_{s_2}, ..., x_{s_n}\}$, where x_{s_i} represents the co-scaled representation of the corresponding patch x_i. The co-scaled representation for each patch x_i captures the cross-scale interactions and dependencies between different patches, enabling a comprehensive understanding of the image content. The co-scaled representation C_S is utilized as input to the co-attention mechanism, which generates an output that captures the interdependencies between the patches. This process enhances the representation by attending to the relationships between different patches. The output of the co-attention mechanism, denoted as $C_A = \{a_1, a_2, \ldots, a_n\}$, consists of attended representations that incorporate global context and cross-scale interactions. These attended representations can then be further processed by subsequent modules in the architecture, such as the encoder and decoder, to extract high-level features and generate outputs such as segmentation masks or classification predictions.

The co-scale and co-attention representations obtained from the segmentation and classification branches of PDLC-ViT are provided as input to the cross-attention module. This module facilitates the integration of these representations, enabling the exchange of information between the segmentation and classification tasks. By leveraging both the local and spatial features captured by the co-scale layer and the global context and cross-scale interactions captured by the co-attention layer, the cross-attention module combines and fuses the representations to create a comprehensive and enriched representation $Z = \{z_1, z_2, ..., z_n\}$. This integrated representation benefits both the segmentation and classification tasks by incorporating localized disease information, and capturing dependencies between different regions. It serves as a basis for generating accurate and detailed outputs, such as segmentation masks and classification predictions.

The shared feature patches are given as input to the Transformer Encoder which has a stack of transformer layers. These layers leverage self-attention mechanisms to capture local and global contextual information from Z. The Transformer encoder output is fed into the segmentation head, which generates a segmentation map representing pixel-level disease segmentation. The classification head receives the outputs of the Transformer encoder and the segmentation head as input. It performs disease classification based on the combined information, providing predictions for the disease classes. The high-level architecture of the PLDC-ViT is illustrated with Fig. 1. In this architecture, the cross-attention mechanism combines convolutional layers and self-attention layers to leverage the strengths of both convolutional and attention mechanisms to capture local and global dependencies effectively, which offers a good trade-off between efficiency and performance.

The Transformer encoder is a critical component of the PDLC-ViT model for plant disease localization and classification. It takes the shared representation from the cross-attention mechanism, and processes it through a series of layers to produce the encoded features. The Transformer encoder is depicted with Fig. 2 and the encoding process is described as below.

Fig. 1 PDLC-ViT high-level architecture



1. *Linear Projection*: The input is first linearly projected. This step transforms the shared representation into a higher dimensional space suitable for the self-attention mechanism.
2. *Positional Encoding*: After the linear projection, positional encodings are added to the input. Since the self-attention mechanism is not inherently sensitive to the sequence order, positional encodings provide necessary information about the position of each element in the sequence.
3. *Multi-head Self-attention*: The core of the Transformer Encoder, the multi-head self-attention layer, allows the model to focus on different parts of the input sequence, capturing complex dependencies. Each "head" computes an attention score, indicating the importance of other parts of the input when encoding a particular piece.
4. *Add & Norm (Addition and Normalization)*: The output from the self-attention mechanism undergoes a residual connection, which adds the input of the layer before the self-attention to its output to help mitigate the vanishing gradient problem. It is then normalized to stabilize the learning process.
5. *Feed-Forward:* This is a fully connected neural network that processes the output of the attention layer separately for each position. It consists of two linear transformations with a Rectified Linear Unit (ReLU) activation function in between.
6. *Add & Norm*: Again, the output from the feed-forward layer is added to its input before the layer (residual connection) and then normalized. This step is similar to the earlier Add & Norm and ensures that the model can learn effectively over many layers without running into training difficulties.

The final output of the Transformer Encoder is a set of encoded features that are passed on to the subsequent parts of the model, such as the segmentation head and the classification head, for further processing.

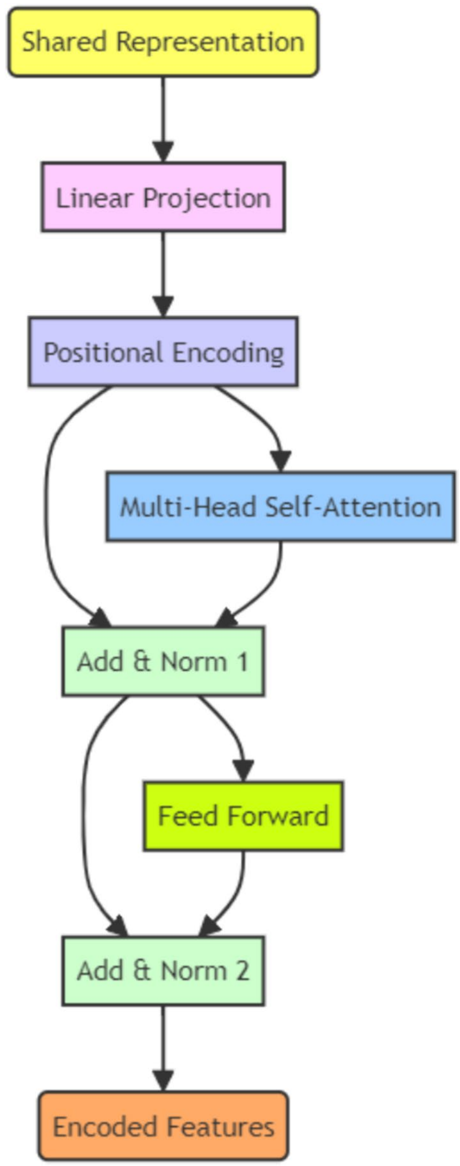


Fig. 2 Transformer encoder

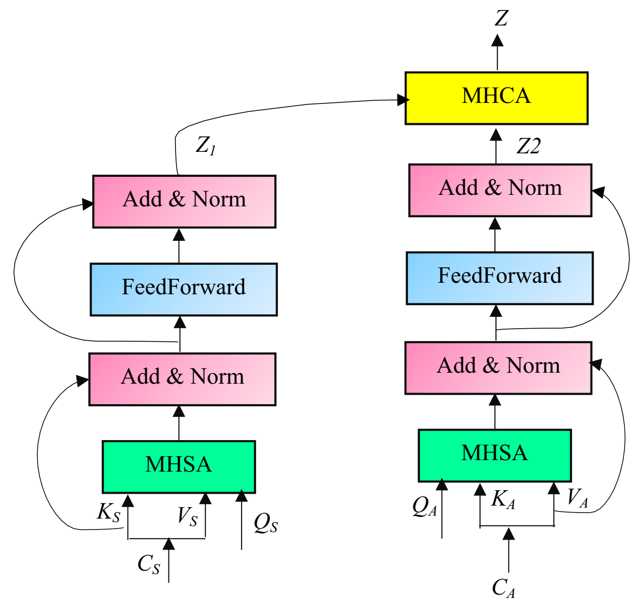


Fig. 3 Cross-attention mechanism

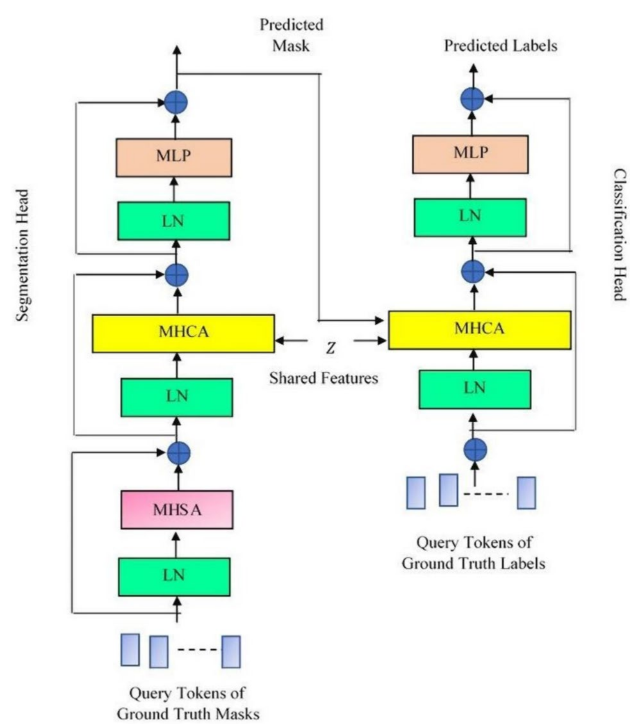


Fig. 4 Segmentation and classification

4.3 Cross-attention Mechanism

The cross-attention mechanism is modeled to create the shared representation Z from the co-scaled and co-attention representations C_S and C_A, respectively. This mechanism is illustrated in Fig. 3 which consists of two parallel branches. Initially, C_S and C_A are transformed into two triples comprising a Query (Q), Key (K), and Value (V) represented as (Q_s, K_s, V_s) and (Q_A, K_A, V_A), respectively. Each triple is given as input to the Multi-Head Self-attention (MHSA) block in each branch which computes the attention scores AS_S and AS_A as in (4) and (5), where d_s and d_A refer to the dimensions of K_s and K_A, respectively

$$AS_S = softmax\frac{(Q_s K_s^T)}{\sqrt{d_s}} V_s \tag{4}$$

$$AS_A = softmax\frac{(Q_A K_A^T)}{\sqrt{d_A}} V_A. \tag{5}$$

In each branch, the attention scores and keys are passed as input to an Add&Norm block which applies an additive operation and Layer Normalization (LN) to enhance the feature representation. The output of this block is given as input to the feed-forward block followed by an Add&Norm block. The outputs Z_1 and Z_2 are generated from the first and second branches, respectively, which incorporate both local and global context, as well as the dependencies between different regions. The shared feature Z is constructed by a Multi-Head Cross-Attention (MHCA) mechanism, which leverages the complementary strengths of both branches to capture comprehensive information and interdependencies between different regions. This integration and fusion of the information from Z_1 and Z_2 result in a shared feature representation denoted as Z.

4.4 Segmentation and Classification

The PDLC-ViT performs segmentation and classification tasks with the segmentation and classification heads as shown in Fig. 4. The segmentation head comprises three blocks within residual connections. The first block constitutes an LN and an MHSA mechanism. The second block comprises an LN block and an MHCA, and the third block comprises the LN and a Multi-Layer Perceptron (MLP). In the segmentation head, the query tokens of the ground truth masks are given as input to the first block to capture the dependencies within the ground truth masks.

The second block of the segmentation head accepts the dependencies captured by the first block, as well as Z, to capture the interdependencies between the shared feature Z and the ground truth. This enables the model to effectively incorporate both the learned representations from the shared feature and the ground truth information. In the third block, the MLP layer, which follows the LN, consists of multiple Fully Connected (FC) layers with the Rectified Linear Unit (ReLU) activation function. This function allows the segmentation head to capture high-level features and relationships within the data, enabling it to extract complex patterns and make accurate predictions for segmentation tasks.

Similarly, the classification head comprises two blocks. The first block constitutes the LN and MHCA. The query tokens of the ground truth class labels corresponding to the segmented image, predicted mask, and the shared feature Z are given as input to this block. The MHCA mechanism allows the classification head to capture the interdependencies between the ground truth class labels and the shared feature representation Z. By attending to these dependencies, the model can effectively incorporate both the learned features from Z and the class label information to make accurate predictions. The second block comprises the MLP layer, which follows the LN, where the MLP consists of a sequence of FC layers with the nonlinear sigmoid activation function. It captures high-level features and relationships within the data, allowing the classification head to extract complex patterns and make accurate predictions.

4.5 PDLC-ViT Training

The training process of the PDLC-ViT model involves optimizing the parameters to minimize the loss between the predicted outputs and the ground truth labels. The training procedure is given as below.

Table 2 PDLC-ViT hyperparameters

Parameter	Value
Initial Learning Rate	0.001
Learning Rate Reduction	Factor of 0.5 every 15 epochs
Validation Loss Threshold	0.001 (indicating convergence)
Optimizer	Adam
Mini-Batch Size	16
Dropout Ratio	0.2 (applied to both encoder and decoder)
L2 Regularization Factor (λ)	0.0005
Data Handling Strategy	Random shuffling during training and testing
Parameter Optimization	Grid search for optimal parameter configuration

Input:

- Image dataset: $I_{dataset} = \{I_1, I_2, \ldots, I_n\}$
- Ground truth masks: $G_{masks} = \{Mask_1, Mask_2, \ldots, Mask_n\}$
- Ground truth class labels: $G\_labels = \{Label\_1, Label\_2, \ldots, Label\_n\}$
- Number of training iterations: N
- Weighting factors for segmentation and classification losses: λ_{cls}, λ_{seg}

Output:

- Trained PDLC-ViT model

1. Initialize the parameters of the PDLC-ViT model, including the segmentation and classification heads.
2. Repeat for i = 1 to N
   i. For each image $I_1 \in I_{dataset}$
   1. Forward Pass:
      - Obtain the shared feature representation Z from the CoaT
      - Segmentation Head:
        i. First Block:
        - Compute the query tokens of the ground truth masks:
          $$Q_{masks} = Embedding(G_{masks})$$
        - Capture the dependencies within the ground truth masks:
          $$H_{masks} = MHSA(LN(Q_{masks}, Z))$$
        ii. Second Block:
        - Capture the dependencies and shared feature:
          $$H_{seg} = MHCA(LN(H_{masks}, Z))$$
        iii. Third Block:
        - Generate the segmentation predictions:
          $$M_{seg} = MLP_{seg}(LN(H_{seg}))$$
      - Classification Head:
        i. First Block:
        - Compute the query tokens of the ground truth class labels:
          $$Q_{labels} = Embedding(G_{labels})$$
        - Capture the dependencies within the ground truth class labels:
          $$H_{labels} = MHCA(LN(Q_{labels}, Z))$$
        ii. Second Block:
        - Capture the dependencies, predicted mask, and shared feature:
          $$Hcls = MHCA(LN(Hlabels, Mseg, Z))$$
        iii. Third Block:
        - Generate the classification predictions:
        - $O_{cls} = MLP_{cls}(LN(H_{cls}))$
   2. Loss Calculation:
      - Compute the segmentation loss:
        $$L_{seg} = SegmentationLoss(M_{seg}, G_{masks})$$
      - Compute the classification loss:
        $$L_{cls} = ClassificationLoss(O_{cls}, G_{labels})$$
      - Compute the total loss:
        $$L_{total} = \lambda_{seg} * L_{seg} + \lambda_{cls} * L_{cls}$$
   3. Backward Pass:
      - Compute gradients of the total loss with respect to the parameters of the segmentation head:
        $$\partial L_{total} / \partial \theta_{seg}$$
      - Compute gradients of the total loss with respect to the parameters of the classification head:
        $$\partial L_{total} / \partial \theta_{cls}$$
      - Update the parameters of the segmentation head using an optimization algorithm:
        $$\theta_{seg} \leftarrow Optimize(\theta_{seg}, \partial L_{total} / \partial \theta_{seg})$$
      - Update the parameters of the classification head using an optimization algorithm:
        $$\theta_{cls} \leftarrow Optimize(\theta_{cls}, \partial L_{total} / \partial \theta_{cls})$$
   4. End For

5 Experimental Results and Discussion

This section describes the empirical results obtained from evaluating the PDLC-ViT model using both the training and testing datasets. The evaluation process incorporates objective evaluation metrics, visual representations, and comparative analyses to thoroughly assess the model's performance. Additionally, an in-depth explainable analysis is performed to gain valuable insights into the behavior of the

Table 3 Plant disease classification types

Classification type	Crop	No. of classes	Target classes
A.	All	2	Healthy, UnHealthy
B.	Apple	4	Scab, BlackRot, Rust, Healthy
C.	Cherry	2	Mildew, Healthy
D.	Corn	4	LeafSpot, Rust, LeafBlight, Healthy
E.	Grape	4	BlackRot, Measles, LeafBlight, Healthy
F.	Peach	2	BacterialSpot, Healthy
G.	Pepper	2	BacterialSpot, Healthy
H.	Potato	3	EarlyBlight, LateBlight, Healthy
I.	Strawberry	2	LeafScorch, Healthy
J.	Tomato	10	Bacterial_Spot,Early_Blight, Healthy, Late Blight, Leaf Curl Virus, Leaf Mold, LeafSpot, SpiderMite, TargetSpot, Mosaic Virus

model, while an ablation study effectively demonstrates the effectiveness of the proposed MTL framework.

5.1 Experimental Setup

The proposed PDLC-ViT is evaluated using the Plant Village dataset as described in Sect. 3.1. The initial learning rate of the model is set to 0.001 which undergoes a gradual reduction by a factor of 0.5 every 15 epochs. Model training is terminated when the validation loss reaches a threshold of 0.001, indicating convergence. The Adam optimizer is employed to optimize the model, utilizing a mini-batch size of 16. Both the encoder and decoder incorporate a dropout ratio of 0.2, while an L2 regularization factor (λ) of 0.0005 is applied across all layers. To prevent overfitting, the training and testing processes employ a random shuffling strategy. Optimal model parameters are determined through a thorough grid search, exploring various parameter configurations.

The number of epochs is not fixed to ensure optimal convergence and prevent overfitting. Instead, early stopping based on the validation loss threshold of 0.001 is employed. This approach allows training to terminate when the model has sufficiently converged, avoiding unnecessary training and overfitting, and ensuring that the model achieves the best performance on the validation set. A thorough grid search is conducted to determine the optimal hyperparameters. This involves systematically varying parameters, such as the learning rate, batch size, dropout ratio, and regularization factor, and evaluating the model's performance on the validation set. It ensures that the chosen hyperparameters yield the best possible performance, contributing to the robustness and reliability of the PDLC-ViT model. Table 2 lists the hyperparameters of the PDLC-ViT model.

5.2 Performance Evaluation

The disease classification and segmentation performance of the PLDC-ViT model is evaluated on the testing dataset with the following objective metrics. Accuracy (ACC), Sensitivity (Sn), Specificity (Sp), Precision (Pre) and F1 metrics defined with Eqs. (6–9) are evaluated from the True-Positive (TP), True-Negative (TN), False-Positive (FP), and False-Negative (FN) values resulting from evaluating the model on the test dataset for classification. These metrics range from 0 to 1, with lower values signifying poor classification, while values around 1 indicate perfect classification. The Area Under the Curve (AUC) is a significant metric that evaluates a classification model's overall performance in distinguishing between positive and negative classes. The ROC curve plots the True-Positive Rate (TPR) against the False-Positive Rate (FPR). A perfect classifier achieves an AUC value of 1, indicating excellent performance, while a random classifier has an AUC of 0.5, representing chance-level performance. The AUC serves as a valuable metric to compare and select the most effective model among different classifiers.

Accuracy is the number of correctly classified samples out of the total number of samples in the dataset, given as in (6). It offers valuable insights into the model's ability in classifying the samples correctly

$$\text{Acc} = \frac{TP + TN}{TP + FP + TN + FN}. \tag{6}$$

Sensitivity or recall measures the model's ability to correctly identify actual positive instances of plant diseases as in (7)

$$\text{Sn} = \frac{TP}{FN + TP}. \tag{7}$$

Specificity (Sp) is a performance metric that quantifies the accurate identification of negative samples in plant disease classification as in (8). It emphasizes the model's ability to minimize FPs and is particularly relevant when the cost of misclassifying non-diseased samples is high

$$\mathrm{Sp} = \frac{TN}{FP + TN}. \tag{8}$$

Precision (Pre) in plant disease classification measures the ability of the model to correctly classify healthy cases. It quantifies the proportion of accurate positive predictions out of all positive classifications made by the model, as given in Eq. (9)

$$\mathrm{Pre} = \frac{TP}{TP + FP}. \tag{9}$$

The F1 score in plant disease classification is a balanced evaluation metric, calculated as the harmonic mean of precision and recall, given as in (10). The F1 score ranges from 0 to 1, with a perfect score of 1 indicating optimal precision and recall, while a score of 0 reflects poor performance in classifying plant diseases

$$\mathrm{F1} = 2 \times \frac{\mathrm{Pre} \times Rec}{\mathrm{Pre} + \mathrm{Rec}}. \tag{10}$$

Similarly, the segmentation performance is evaluated with the Dice and Intersection over Union (IoU) metrics. In the context of disease localization in leaf lesion segmentation, the Dice coefficient evaluates the overlap between the ground truth G and the predicted mask P, as defined in Eq. (11).

It specifically measures the precision of lesion localization by quantifying the degree of overlap between the two masks. The IoU metric assesses the accuracy of lesion localization by calculating the ratio of the intersection to the union of G and P, as expressed in Eq. (12). Both the Dice and IoU values range from 0 to 1, with a score of 1 indicating a perfect localization of lesions

$$Dice = \frac{2|P \cap G|}{|P| + |G|} \tag{11}$$

$$IoU = \frac{|P \cap G|}{|P \cup G|}. \tag{12}$$

In addition to the above, the Mean Average Precision (MAP) and Mean Average Recall (MAR) metrics are used in this research. MAP is defined as the mean of the Average Precision (AP) scores for all classes across multiple thresholds as in (13). AP for a single class is calculated by integrating the precision–recall values as in (14). Similarly, MAR is the mean of the Average Recall (AR) scores for all classes across multiple thresholds as in (15). AR for a single class is calculated by integrating the recall–precision values as in (16)

$$\mathrm{MAP} = \frac{1}{Q} \sum_{q=1}^{Q} AP_q \tag{13}$$

$$AP = \sum_{n} \left(R_n - R_{n-1}\right) P_n \tag{14}$$

$$MAR = \frac{1}{Q} \sum_{q=1}^{Q} AR_q \tag{15}$$

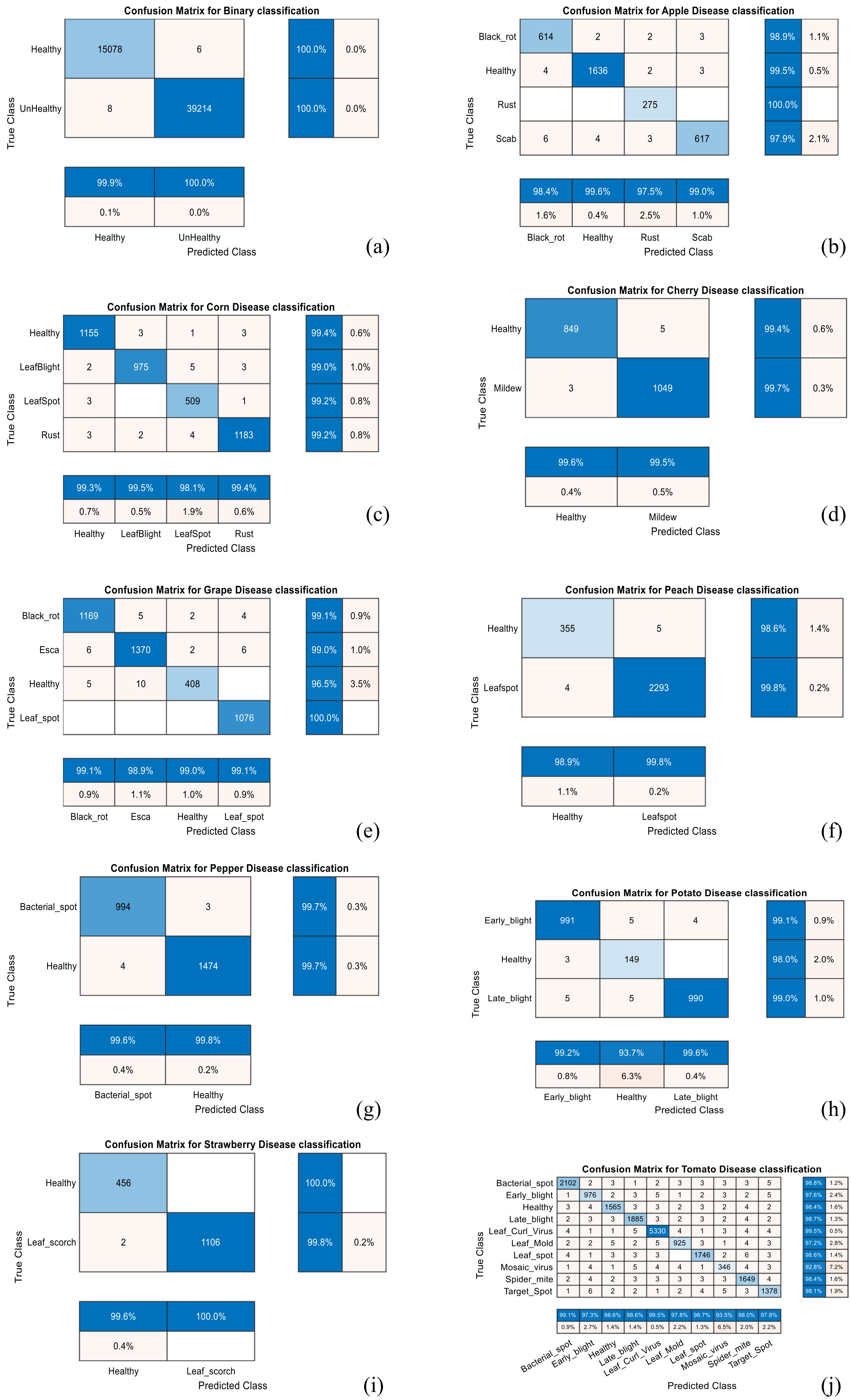
$$AR = \sum_{n} \left(P_n - P_{n-1}\right) R_n. \tag{16}$$

5.3 Disease Classification

The classification performance of the model is evaluated under 10 cases as in Table 3. The performance metrics are given in Table 4 for each classification type. It is seen that

Table 4 Plant disease classification: performance metrics

Classification Type	Accuracy	Sensitivity	Specificity	Precision	F1 Score	AUC
A.	**0.9997**	**0.9996**	**0.9998**	**0.9995**	**0.9995**	**0.9993**
B.	0.9909	0.9907	0.9968	0.9865	0.9885	0.9861
C.	0.9958	0.9941	0.9971	**0.9965**	0.9953	0.9940
D.	0.9922	0.9921	0.9974	0.9907	0.9914	0.9903
E.	0.9902	0.9863	0.9965	0.9902	0.9882	0.9860
F.	0.9966	0.9861	0.9983	0.9889	0.9875	0.9860
G.	0.9972	0.9970	0.9973	0.9960	0.9965	0.9961
H.	0.9898	0.9871	0.9949	0.9750	0.9809	0.9821
I.	**0.9987**	**1.0**	0.9982	0.9956	**0.9978**	**0.9971**
J.	0.9858	0.9781	**0.9984**	0.9788	0.9784	0.9781


Confusion Matrix for Binary classification
Healthy
UnHealthy
True Class
Predicted Class
(a)
Confusion Matrix for Apple Disease classification
Black_rot
Healthy
Rust
Scab
True Class
Predicted Class
(b)
Confusion Matrix for Corn Disease classification
Healthy
LeafBlight
LeafSpot
Rust
True Class
Predicted Class
(c)
Confusion Matrix for Cherry Disease classification
Healthy
Mildew
True Class
Predicted Class
(d)
Confusion Matrix for Grape Disease classification
Black_rot
Esca
Healthy
Leaf_spot
True Class
Predicted Class
(e)
Confusion Matrix for Peach Disease classification
Healthy
Leafspot
True Class
Predicted Class
(f)
Confusion Matrix for Pepper Disease classification
Bacterial_spot
Healthy
True Class
Predicted Class
(g)
Confusion Matrix for Potato Disease classification
Early_blight
Healthy
Late_blight
True Class
Predicted Class
(h)
Confusion Matrix for Strawberry Disease classification
Healthy
Leaf_scorch
True Class
Predicted Class
(i)
Confusion Matrix for Tomato Disease classification
Bacterial_spot
Early_blight
Healthy
Late_blight
Leaf_Curl_Virus
Leaf_Mold
Leaf_spot
Mosaic_virus
Spider_mite
Target_Spot
True Class
Predicted Class
(j)

◂**Fig. 5** Confusion matrices for plant disease classification: (**a**) unhealthy vs healthy; (**b**) apple; (**c**) cherry; (**d**) corn; (**e**) grape; (**f**) peach; (**g**) pepper; (**h**) potato; (**i**) strawberry; (**j**) tomato

the model demonstrates best performance metrics for the binary classification of healthy and unhealthy plants, irrespective of the species. However, the model exhibits mixed detection performances for different crops. This can be attributed to the diversities in different plant species and imbalanced classes.

The confusion matrices illustrating the correct and misclassifications are illustrated with Fig. 5 to understand the ability of the model to discern the classes. It is evidenced that the model correctly classifies the Rust disease in Apples with no misclassifications, while a few incorrect classifications are observed with other classes. Similarly, in Cherry, while 5 Healthy cases are misclassified as Mildew, 3 Mildew cases are classified as Healthy. The model exhibits > 99% of all performance metrics in the multiclass classification of Corn diseases, exhibiting best values for LeafSpot. Similarly, in the classification of Grape diseases, the model shows best performance in the detection of LeafSpots with no misclassifications. In the binary classification of Peach and Pepper, the model discerns the Healthy cases from BacterialSpots with an accuracy of 0.9966 and 0.9972, respectively. In the three-class classification of Potato, the model shows a reasonable accuracy of 0.9898 and specificity of 0.9949. In the binary classification of Strawberry, the model demonstrates the highest classification accuracy of 0.9987 next to the binary classification accuracy and the highest sensitivity of 1.0. Finally, in the 10-class tomato classification, the model exhibits the classification accuracy of 0.9858, lowest compared to all categories. On the whole, it is observed that the model demonstrates exceptional performance in certain specific scenarios, but its performance shows variation when dealing with complex and diverse plant disease classes. These variations can be attributed to factors such as dataset diversity, class imbalances, and the complexity of different disease patterns. To achieve better performance across all classes, incorporation of balanced datasets and the adoption of tailored strategies to address class-specific challenges are necessary. Additionally, further fine-tuning and optimization of the model architecture could lead to enhanced robustness and improved generalization capabilities, enabling the model to effectively handle a broader range of plant diseases.

In this research, MAP and MAR are evaluated under three different thresholds: 0.3, 0.5, and 0.7. These thresholds are chosen to provide a comprehensive evaluation of the PDLC-ViT model's performance across varying levels of sensitivity and specificity. MAP assesses the model's precision across these thresholds, with a high value indicating consistent true-positive identification and minimal false positives, which is crucial for precise plant disease detection. MAR measures the model's ability to identify all true positives across these thresholds, essential for preventing disease spread and ensuring comprehensive crop health monitoring. The high MAP of 99.18% and MAR of 99.11% underscore the model's exceptional precision and recall, highlighting its robust and balanced performance in accurately detecting plant diseases. This ensures comprehensive monitoring and early intervention, making PDLC-ViT a reliable and advanced tool for agricultural applications, setting a new benchmark in plant disease detection.

5.4 Disease Localization

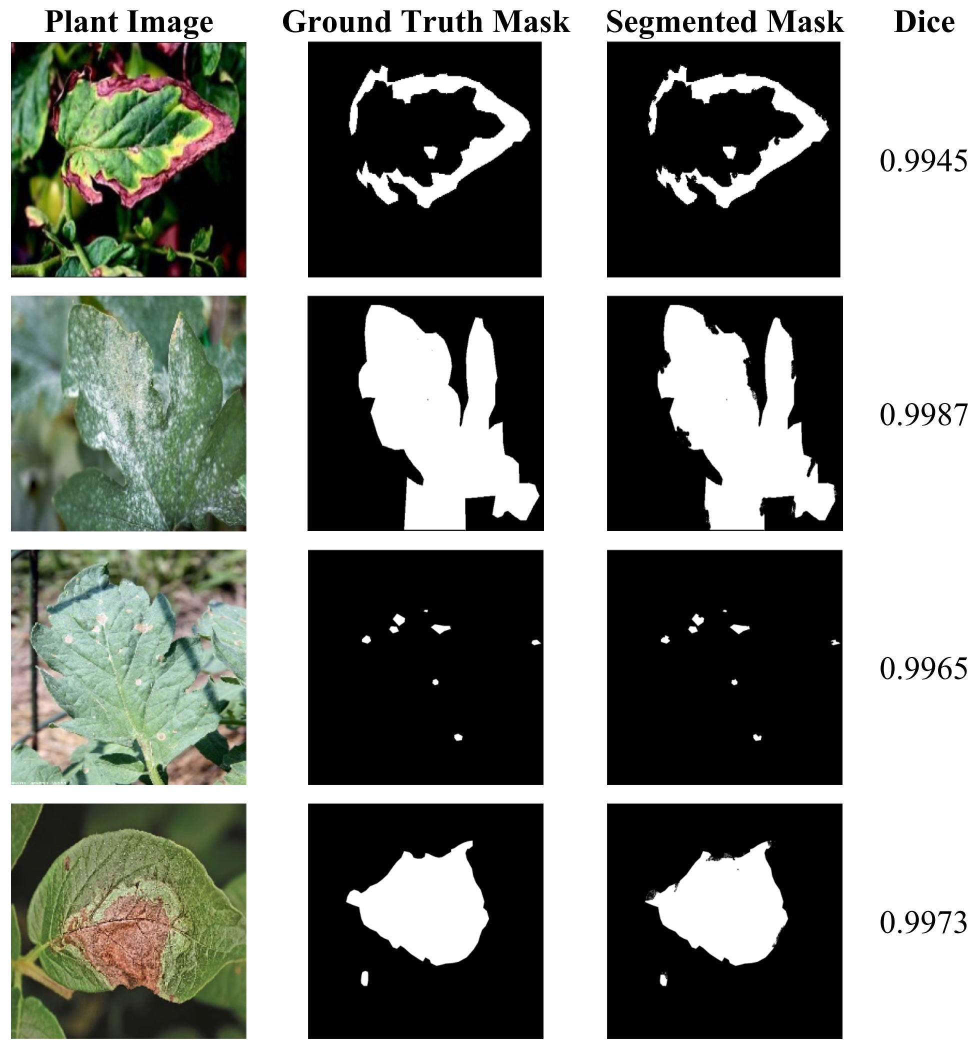
The ability of the model to segment the infected regions for disease localization is evaluated with the Plantdoc dataset using the segmentation metrics. These results are presented in Fig. 6 with the leaf images, ground truth, and segmented masks with objective metrics. It is seen that the predicted masks match with the ground truth labels with high Dice values. The image demonstrates the exceptional performance of the PDLC-ViT model in segmenting various plant diseases, as indicated by the high Dice coefficients for each disease type. The model accurately segments leaves affected by Anthracnose, Powdery Mildew, Early Blight, and Late Blight, with Dice coefficients of 0.9945, 0.9987, 0.9965, and 0.9973, respectively. These diseases exhibit distinct characteristics: Anthracnose with dark, sunken lesions; Powdery Mildew with widespread white, powdery spots; Early Blight with small, scattered brown lesions; and Late Blight with large, water-soaked lesions bordered by pale green tissue. PDLC-ViT effectively captures these varied disease patterns, showcasing its precision and robustness. The model's ability to accurately identify and segment both dense and sparse disease manifestations underscores its potential as a reliable tool for early disease detection and comprehensive crop management, ultimately enhancing agricultural productivity and sustainability. Further, the mean Dice and mean IoU values of 0.9921 and 0.9973, respectively, are achieved on the test dataset.

5.5 Comparative Analysis

This section compares the proposed PDLC-ViT with representative models with respect to the model characteristics and performance metrics in Table 5. Among these models, the Multi-input Network [21], CMTL[22], and MTLSegFormer [33] are MTL-based models similar to PDLC-ViT, and the others are ViT-based models, designed for independent tasks.

However, the MTL-based models also differ by the nature of tasks for which they are built. While the

Fig. 6 Disease localization



Multi-input Network and CMTL focus on plant and species identification, MTLSegFormer is meant for semantic segmentation from learned features of other tasks.

It is understood that the proposed PDLIC-ViT is the only model designed for simultaneous segmentation and classification by sharing the features. By integrating segmentation and classification tasks, PDLC-ViT captures intricate relationships between disease regions and classes, resulting in more accurate predictions. This diversity across these models does not allow direct comparison between these models. However, the results presented in Table 5 show that PDLIC-ViT exhibits the highest classification accuracy of 99.97%, compared to the rest. It surpasses the 99.86% classification accuracy of MSCVT [26], which extracts features at multiple scales, by 0.11%. Further, PDLC-ViT outperforms MTLSegFormer by a significant margin of 19% in IoU. Analysis of the model characteristics and the quantitative metrics from this table show that PDLC-ViT is unique in the context of model design and tasks trained, and superior in performance compared to the state-of-the-art.

5.6 Ablation Study

Ablation study is conducted on the PDLC-ViT model to evaluate the individual contributions of different components and mechanisms in the proposed architecture. It serves as an essential analysis to understand the significance of each element and how they collectively impact the model's performance. By systematically removing specific components, such as the Co-attention and Co-scale mechanisms, and the segmentation head, insights into their effects on the plant disease classification tasks is obtained. The results of ablation study are presented in Table 6 for binary and multiclass classifications for three cases along with the baseline.

The results show that the Co-attention and Co-Scale mechanisms are crucial for capturing dependencies and interactions between tasks, while the Segmentation Head is

Table 5 Comparative analysis

Model	Dataset	Performance metrics	Model characteristics	Limitations
Multi-input Network [21] Keceli et al. (2022)	Plant Village	Acc: 98% Sp: 99% Sn: 96% Pre: 98% F1: 97%	Model employs multi-task learning, shared representations, and a multi-input network to automate plant species and disease identification	The model's performance may be affected if the pre-trained deep model used for feature extraction is not well suited for the specific plant species and diseases being targeted
CMTL[22] Lee et al. (2021)	Plant Village	Acc: 94.52%	Jointly Learns the species and disease classes simultaneously	Model depends on a large labeled dataset for both host species and disease characteristics. Limited or imbalanced data may result in reduced identification accuracy
ViT [24] Thai et al. (2021)	Cassava Leaf Disease Dataset	1% > CNN models	Deployed in Raspberry Pi 4 for automated disease detection	Model cannot effectively handle small or subtle patterns in cassava leaf images, particularly characteristic of certain diseases
PlantXViT [25] Thakur et al. (2022)	Plant Village	Acc: 98.86%	Explainable light-weight model based on CNN and ViT	Additional evaluation on a broader range of crops and diseases is necessary to assess the model's generalization capabilities
MSCVT [26] Zhu et al. (2023)	Plant Village	Acc: 99.86%	The fusion of local and global features at both shallow and deep levels of the model is facilitated by integrating multiscale convolution and self-attention mechanisms	The use of inverted residual blocks to reduce the number of parameters may lead to overfitting if not adequately regularized
ConvViT [27] Li et al. (2023)	Private apple leaf disease dataset	Acc: 96.85%	Combines convolutional and Transformer structures to leverage their respective strengths	Cannot handle leaf images with diverse backgrounds
SLViT [28] Li et al. (2023)	Plant Village	Acc: 98.84%	Employs Transformer plugins in CNN architecture	SLViT is trained on the Plant Village dataset and fine-tuned on the self-created SLD10k dataset. The difference in data distribution between the two datasets could introduce dataset bias
MTLSegFormer [33] Goncalves et al. (2023)	Defoliation and Leaf datasets	Leaf F1: 0.9869 IoU: 0.9743 Defoliation F1: 0.8877 IoU: 0.8048	MTL-based ViT for semantic segmentation	Model may not generalize well to completely new or unrelated tasks
PDLC-ViT (Proposed)	Plant Village Plantdoc	Acc::99.97% Mean Dice:0.9921 Mean IoU:0.9973	Co-scale, co-attention, and cross-attention-based ViT for joint segmentation and classification	The model has not been tested end–end with a single dataset

Table 6 Performance metrics under ablation study

Model	Binary classification					Multiclass classification				
	Acc	Sn	Sp	Pre	F1	Mean Acc	Mean Sn	Mean Sp	Mean Pre	Mean F1
PDLC-ViT (Baseline)	**0.9997**	**0.9996**	**0.9998**	**0.9995**	**0.9995**	**0.9937**	**0.9911**	**0.9975**	**0.9898**	**0.9904**
PDLC-ViT-without co-attention & co-scale	0.9497	0.9496	0.9498	0.9495	0.9495	0.9241	0.9217	0.9276	0.9205	0.9211
PDLC-ViT without Segmentation Head	0.8897	0.8896	0.8898	0.8896	0.8896	0.8546	0.8524	0.8578	0.8512	0.8517

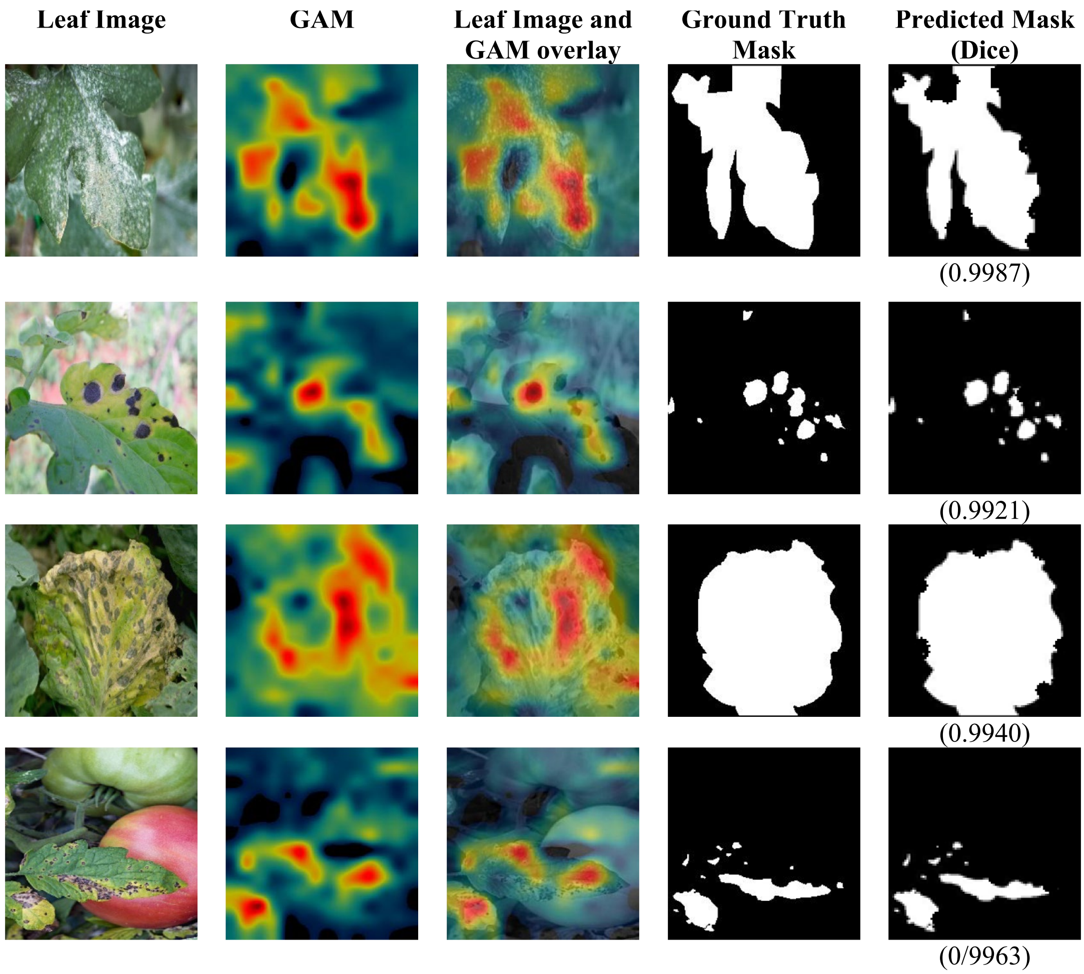


Fig. 7 Explainable analysis of PDC-ViT

essential for accurate segmentation tasks to improve classification performance. When these components are removed, the model's performance is significantly affected, with lower accuracy, sensitivity, specificity, precision, and F1 score, as observed in both binary and multiclass classifications compared to the baseline. The study emphasizes the importance of the proposed architecture in the PDLC-ViT model for simultaneous segmentation and classification tasks in plant disease detection. Understanding the impact of these components can guide further optimization of the model, leading to improved performance for real-world agricultural applications.

5.7 Explainable Analysis

The process of explainable analysis of deep learning models, aimed at discovering the underlying decision process of the model, is considered critical for adoption by non-expert users, particularly in medical image classification and segmentation tasks. In this research, the Explainable Artificial Intelligence (XAI) approach is employed to understand the behavior of PDLC-ViT using attention maps. The Gradient-weighted Class Activation Mapping (Grad-CAM) [36] approach is utilized to generate Gradient Activation Maps (GAMs) for analyzing PDLC-ViT and explaining its decisions. The model's interpretability is established through these GAMs, which provide intuitive visualizations and insights into its behavior. By showcasing image slices, attention maps, ground truth, and predicted masks in Fig. 7 for selected test samples, the research gains a better understanding of PDLC-ViT's decision-making process, making it more accessible for precise disease identification tasks.

The model is capable of localizing the infections with a good degree of accuracy as evident from the visual similarities between the ground truth and the predicted masks. Precise segmentation of the infected regions by the segmentation head facilitates the classification head to discriminate the disease types.

5.8 Discussion

Empirical evaluations of the PDLC-ViT on the Plant Village and Plantdoc datasets demonstrate its superior performance compared to ViT-based classifier models and the MTLSegformer. The success of PDLC-ViT is attributed to its unique approach of joint learning, where the model effectively leverages shared features captured from the co-attention and co-scaled representations of the input image. By integrating both global and local dependencies through co-attention mechanisms and hierarchical feature extraction via co-scaled representations, PDLC-ViT achieves a comprehensive understanding of the input image, enabling more accurate disease identification and segmentation. The model's ability to jointly learn and utilize information from both tasks enhances its performance and generalization capabilities, outperforming other approaches that treat segmentation and classification as separate tasks.

Explainable analysis shows that PDLC-ViT is capable of localizing the subtle infected regions distributed across the leaves. By sharing features between these tasks, the model can capture important patterns and relationships that contribute to accurate disease localization. The importance of different regions based on their relevance to the specific task at hand is weighed by the model and learned through the utilization of attention mechanisms. This enables PDLC-ViT to give higher importance to disease-specific regions during segmentation, leading to more accurate localization of infected areas which subsequently improves the classification performance.

Moreover, the efficient feature sharing mechanism reduces redundancy and optimizes resource utilization, making PDLC-ViT well suited for real-world applications in smart agriculture and disease detection. The results clearly indicate that PDLC-ViT represents a promising advancement in MTL based on a ViT for plant disease detection, setting a new standard in the field and offering significant benefits in terms of accuracy, robustness, and efficiency.

However, this model is trained with different datasets on the segmentation and classification tasks due to the lack of an integral dataset for end–end training, which can lead to domain shifts and performance discrepancies. To address this challenge, one possible solution is to create a unified dataset that includes both segmentation and classification annotations. By training the model on a combined dataset, it can learn to better integrate the information from both tasks and achieve more consistent performance. Alternatively, data augmentation and domain adaptation techniques can be employed to mitigate the domain shift and improve the model's generalization capabilities across different datasets. These approaches can help enhance the overall performance and reliability of the PDLC-ViT model in plant disease detection tasks.

6 Conclusions

This paper presents the PDLC-ViT model based on joint learning of segmentation and classification tasks for plant disease classification. This model is tested on the publicly available Plant Village and Plantdoc datasets demonstrating superior binary and multiclass classification performances. The model exhibits desirable detection performances for a wide variety of crops, such as Apple, Corn, Grapes, Peach, Potato, Tomato, etc. Further, the localization ability of the model facilitates the detection of diseases at the early stages. The ability of the model to learn from shared features improves the classification accuracy of the model. PDLC-ViT can be used as a baseline in the disease detection of several other agricultural species, reducing the loss of yield and enhancing crop health.

The PLDC-ViT model can be customized to potential applications, including pest detection and monitoring, stress analysis due to environmental factors, and crop growth stage monitoring. By detecting various pests, the model can provide early warnings and enable timely pest management, reducing crop losses and minimizing pesticide use. It can also monitor plant responses to abiotic stresses, such as drought, salinity, and nutrient deficiencies, allowing farmers to take early action to ensure better yields. Additionally, the model can be used to accurately identify different crop growth stages to optimize irrigation, nutrient application, and harvesting, contributing to precision agriculture and resource efficiency.

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Data Availability Plant village dataset: https://data.mendeley.com/datasets/tywbtsjrjv/1.

Declarations

Conflict of interest The authors declare no conflicts of interest related to the research presented in this manuscript.

Ethics Approval The research is original and has not been submitted elsewhere. Ethical guidelines and data-sharing policies have been adhered to.

Consent to Publish We confirm that the submission is original, and has not been published elsewhere, and all co-authors agree with the submission.

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